

THEORY AND METHODS

A CLASS OF ESTIMATORS IN TWO-STAGE SAMPLING WITH VARYING PROBABILITIES

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Summary: This paper introduces a general class of estimators for the finite population total, using a single auxiliary variable in a two-stage sampling with varying probabilities.

1. Introduction

Let the first-stage units (fsu) of a population be labelled $i = 1, 2, \dots, N$ such that the i -th fsu contains M_i second-stage units (ssu) and $M = \sum_{i=1}^N M_i$. Let Y_i and X_i be the totals of the i -th fsu in respect of the study variable y and the auxiliary variable x respectively. Suppose that our aim is an estimation of the overall total $Y = \sum_{i=1}^N Y_i$ of y -values when the overall total $X = \sum_{i=1}^N X_i$ of the x -values is known. A sample s of n fsu's is selected according to any design with π_i and π_{ij} as the known first and second order inclusion probabilities. Note that for any fixed sample size n , π_i and π_{ij} are functions of n . The selected fsu's are subsampled independently with suitable selection probabilities in the second stage.

Let E_1, E_2 ($V_1, V_2; Cov_1, Cov_2$) denote the expectations (variances, covariances) over repeated sampling in the first and second stages; by

$E(V, Cov)$ we denote overall expectation (variance, covariance). When the unit i is selected, it is assumed that from sampling in the second stage estimators t_{iy} and t_{ix} respectively for Y_i and X_i are available such that $E_2(t_{iy}) = Y_i$, $E_2(t_{ix}) = X_i$, $V_2(t_{iy}) = \sigma_{iy}^2$, $V_2(t_{ix}) = \sigma_{ix}^2$ and $Cov_2(t_{iy}, t_{ix}) = \sigma_{iyx}$.

Given the sample s of fsu's we define the estimators $t_y = \sum_{i \in s} t_{iy}/\pi_i$ and $t_x = \sum_{i \in s} t_{ix}/\pi_i$ such that

$$E(t_y) = Y, E(t_x) = X, V(t_y) = \sigma_y^2 + \sum_{i=1}^N \sigma_{iy}^2/\pi_i,$$

$$V(t_x) = \sigma_x^2 + \sum_{i=1}^N \sigma_{ix}^2/\pi_i \text{ and } Cov(t_y, t_x) = \sigma_{yx} + \sum_{i=1}^N \sigma_{iyx}/\pi_i$$

where

$$\sigma_y^2 = \frac{1}{2} \sum_{i \neq j=1}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i}{\pi_i} - \frac{Y_j}{\pi_j} \right)^2,$$

$$\sigma_x^2 = \frac{1}{2} \sum_{i \neq j=1}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{X_i}{\pi_i} - \frac{X_j}{\pi_j} \right)^2,$$

and

$$\sigma_{yx} = \frac{1}{2} \sum_{i \neq j=1}^N (\pi_i \pi_j - \pi_{ij}) \left(\frac{Y_i}{\pi_i} - \frac{X_i}{\pi_i} \right) \left(\frac{Y_j}{\pi_j} - \frac{X_j}{\pi_j} \right).$$

Srivastava (1980) considered $\hat{Y}_G = h(t_y, t_x)$ as a class of estimators for Y , where $h(t_y, t_x)$ is a function of t_y and t_x admitting certain regularity conditions such that $h(Y, X) = Y$. This class makes use of a pre-assigned value of X . But, in practice, very often the auxiliary variable values X_i are known for the fsu's. Hansen *et al.* (1953) and Smith (1969) gave brief discussions of this type of situations.

In this note, following Srivastava (1980), a general class of estimators for Y has been developed under the assumption that both X and the X_i 's for the selected fsu's, are known.

2. The class of estimators

For the i -th selected fsu, let us define a class of estimators for Y_i by

$$\hat{Y}_i = f_{2i}(t_{iy}, t_{ix}),$$

where $f_{2i}(t_{iy}, t_{ix})$ is a known function of t_{iy} and t_{ix} , which may contain X_i but is independent of Y_i , such that $f_{2i}(Y_i, X_i) = Y_i$.

Let $t'_y = \sum_{i \in s} \hat{Y}_i / \pi_i$, $t'_x = \sum_{i \in s} \frac{X_i}{\pi_i}$ and $f_1(t'_y, t'_x)$ be a function of t'_y and t'_x which may contain X but is independent of Y , such that $f_1(Y, X) = Y$.

Further, for the given s , let us assume that

- (i) $(t_{iy}, t_{ix}) \forall i$, and (t'_y, t'_x) take values in a bounded, closed convex subset, S , of two-dimensional real space containing the points (Y_i, X_i) and (Y, X) , and
- (ii) the functions f_{2i} and f_1 are continuous having first and second order partial derivatives which are continuous in subset S .

These conditions are, however, similar to those of Srivastava (1980) for justifying $\hat{Y}_S$ to be a class of estimators for Y .

We now consider a general class of estimators of Y defined by

$$\hat{Y}_C = f_1(t'_y, t'_x). \quad (2.1)$$

Expanding $\hat{Y}_i = f_{2i}(t_{iy}, t_{ix})$ about (Y_i, X_i) in a second order Taylor's series, it can be observed that to a first order of approximation $E_2(\hat{Y}_i) \simeq Y_i$ with asymptotic variance

$$V_2(\hat{Y}_i) = \sigma_{iy}^2 + 2f_{22i}(Y_i, X_i)\sigma_{iyx} + f_{22i}^2(Y_i, X_i)\sigma_{ix}^2, \quad (2.2)$$

where $f_{22i}(Y_i, X_i)$ is the value of the first order partial derivative of f_{2i} w.r.t. t_{ix} at the point (Y_i, X_i) .

Further, expanding $f_1(t'_y, t'_x)$ about (Y, X) in a second order Taylor's series, the asymptotic variance of $\hat{Y}_C$ is

$$V(\hat{Y}_C) = V(t'_y) + 2f_{12}(Y, X)\text{Cov}(t'_y, t'_x) + f_{12}^2(Y, X)V(t'_x) \quad (2.3)$$

where $f_{12}(Y, X)$ is the value of the first order partial derivative of f_1 w.r.t. t'_x at (Y, X) .

We now have $V(t'_y) = V_1E_2(t'_y) + E_1V_2(t'_y)$ which on simplification reduces to

$$V(t'_y) = \sigma_y^2 + \sum_{i=1}^N V_2(\hat{Y}_i)/\pi_i.$$

Similarly, using conditional expectations we may also have $\text{Cov}(t'_y, t'_x) \simeq \sigma_{yx}$ and $V(t'_x) = \sigma_x^2$.

Finally, the formula for the asymptotic variance of $\hat{Y}_C$ is obtained as

$$\begin{aligned} V(\hat{Y}_C) &= \sigma_y^2 + 2f_{12}(Y, X)\sigma_{yx} + f_{12}^2(Y, X)\sigma_x^2 \\ &\quad + \sum_{i=1}^N [\sigma_{iy}^2 + 2f_{22i}(Y_i, X_i)\sigma_{iyx} + f_{22i}^2(Y_i, X_i)\sigma_{ix}^2] / \pi_i \end{aligned} \quad (2.4)$$

3. Some observations and remarks

3.1 In general, $\hat{Y}_C$ and $\hat{Y}_S$ represent two different classes of estimators. However, they reduce to an equivalent class when

$$f_{2i}(t_{iy}, t_{ix}) = t_{iy}, X_i = t_{ix}, i = 1, 2, \dots, N \text{ and } f_1(.,.) = h(.,.).$$

The asymptotic variance of $\hat{Y}_S$ is

$$\begin{aligned} V(\hat{Y}_S) &= \sigma_y^2 + 2h_2(Y, X)\sigma_{yx} + h_2^2(Y, X)\sigma_x^2 \\ &+ \sum_{i=1}^N [\sigma_{iy}^2 + 2h_2(Y, X)\sigma_{iyx} + h_2^2(Y, X)\sigma_{ix}^2] / \pi_i \end{aligned} \quad (3.1)$$

where $h_2(Y, X)$ is the value of the first order partial derivative of h w.r.t. t_x at (Y, X) .

From (2.4) and (3.1) it follows that $V(\hat{Y}_C) \leq V(\hat{Y}_S)$ if

$$(i) |h_2(Y, X) + \beta_{yx}| \geq |f_{12}(Y, X) + \beta_{yx}|$$

and

$$(ii) |h_2(Y, X) + \beta_{iyx}| \geq |f_{22i}(Y_i, X_i) + \beta_{iyx}| \quad \forall i,$$

where $\beta_{yx} = \sigma_{yx}/\sigma_x^2$ and $\beta_{iyx} = \sigma_{iyx}/\sigma_{ix}^2$.

If $f_1(.,.) = h(.,.)$, then (ii) will be the only sufficient condition for $\hat{Y}_C$ to be more precise than $\hat{Y}_S$.

3.2 The variance of $\hat{Y}_C$ given in (2.4) is sought to be minimised subject to $f_{12}(Y, X) = -\beta_{yx}$ and $f_{22i}(Y_i, X_i) = -\beta_{iyx} \quad \forall i$, and the resulting minimum asymptotic variance is

$$\min V(\hat{Y}_C) = \sigma_y^2(1 - \lambda_{yx}^2) + \sum_{i=1}^N \sigma_{iy}^2(1 - \lambda_{iyx}^2)/\pi_i, \quad (3.2)$$

where $\lambda_{yx} = \sigma_{yx}/\sigma_y\sigma_x$ and $\lambda_{iyx} = \sigma_{iyx}/\sigma_{iy}\sigma_{ix}$.

The estimator attaining this minimum variance is a regression-type estimator of the form

$$t_{RG} = \sum_{i \in s} [t_{iy} - \beta_{iyx}(t_{ix} - X_i)]/\pi_i - \beta_{yx}(t'_x - X),$$

which can be transformed to the regression-type estimator considered by Sahoo (1987) on adopting SRSWOR design in different stages.

3.3 Assume that the fsu's are selected with replacement such that $P_i = M_i/M$ is the selection probability assigned to the i -th fsu ($i = 1, 2, \dots, N$) and then a subsample of m_i ssu is selected from the i -th selected fsu by SRSWOR. Let $\bar{y}_i$ and $\bar{x}_i$ be the sample means of y - and x -values in the i -th selected fsu based on m_i units. Then, we have $t_{iy} = M_i\bar{y}_i$, $t_{ix} = M_i\bar{x}_i$, $t_y = \frac{1}{n} \sum_{i \in s} M_i\bar{y}_i/P_i$, $t_x = \frac{1}{n} \sum_{i \in s} M_i\bar{x}_i/P_i$, $t'_y = \frac{1}{n} \sum_{i \in s} \hat{Y}_i/P_i$ and $t'_x = \frac{1}{n} \sum_{i \in s} X_i/P_i$. For this sampling model we obtain

$$\begin{aligned} V(\hat{Y}_C) &= \frac{1}{n} [\sigma_{by}^2 + 2f_{12}(Y, X)\sigma_{byx} + f_{12}^2(Y, X)\sigma_{bx}^2] \\ &\quad + \frac{1}{n} \sum_{i=1}^N \frac{M_i^2}{P_i} \left(\frac{1}{m_i} - \frac{1}{M_i} \right) \\ &\quad \cdot [S_{iy}^2 + 2f_{22i}(Y_i, X_i)S_{iyx} + f_{22i}^2(Y_i, X_i)S_{ix}^2], \end{aligned}$$

where

$$\sigma_{byx} = \sum_{i=1}^N P_i \left(\frac{Y_i}{P_i} - Y \right) \left(\frac{Y_i}{P_i} - X \right)$$

$$S_{iyx} = \frac{1}{M_i - 1} \sum_{k=1}^{M_i} \left(y_{ik} - \frac{Y_i}{M_i} \right) \left(x_{ik} - \frac{X_i}{M_i} \right),$$

$y_{ik}(x_{ik})$ is the y -value (x -value) for the k -th ssu in the i -th fsu; corresponding expressions for σ_{by}^2 , σ_{bx}^2 , S_{iy}^2 and S_{ix}^2 can be obtained using $y_{ik} = x_{ik}$.

3.4 The suggested class can easily be reduced to a large number of estimators including those which appear in the literature for a two-stage sampling model with proper choices of the functions f_1 and f_{2i} . When the allowable design is SRSWOR in both the stages, the following estimators will be the special cases arising from the general formula (2.1):

(i) Simple unbiased estimator: $\hat{Y} = \frac{N}{n} \sum_{i \in s} M_i \bar{y}_i$

(ii) Smith's (1969) ratio estimator: $\hat{Y}_{SR} = \frac{\sum_{i \in s} M_i \bar{y}_i}{\sum_{i \in s} X_i} X$

(iii) Murthy's (1967) chain ratio estimator considered by Sahoo and Swain (1983):

$$\hat{Y}_{CR} = \frac{\sum_{i \in s} \frac{\bar{y}_i}{\bar{x}_i} X_i}{\sum_{i \in s} X_i} X$$

(iv) Sahoo and Swain's (1986) chain product estimator:

$$\hat{Y}_{CP} = \frac{\left[\frac{N}{n} \sum_{i \in s} \frac{(M_i \bar{y}_i)(M_i \bar{x}_i)}{X_i} \right] \left[\frac{N}{n} \sum_{i \in s} X_i \right]}{X}$$

It will also be interesting to verify that the estimators like

$$\hat{Y}_1 = \frac{N}{n} \sum_{i \in s} M_i \bar{y}_i \left(\frac{X_i}{M_i \bar{x}_i} \right)^{\alpha_{2i}} \left[\frac{n}{N} \frac{X}{\sum_{i \in s} X_i} \right]^{\alpha_1}$$

$$\hat{Y}_2 = \frac{N}{n} \sum_{i \in s} M_i \bar{y}_i \frac{X_i}{\alpha_{2i} M_i \bar{x}_i + (1 - \alpha_{2i}) X_i} \cdot \frac{X}{\frac{N}{n} \alpha_1 \sum_{i \in s} X_i + (1 - \alpha_1) X}$$

$$\hat{Y}_3 = \frac{N}{n} \sum_{i \in s} [M_i \bar{y}_i + d_i (M_i \bar{x}_i - X_i)] \left[\frac{n}{N} \frac{X}{\sum_{i \in s} X_i} \right]^\alpha$$

$$\hat{Y}_4 = \frac{N}{n} \sum_{i \in s} \left(\frac{M_i \bar{y}_i}{M_i \bar{x}_i} \right) X_i + d \left(\frac{N}{n} \sum_{i \in s} X_i - X \right)$$

$$\hat{Y}_5 = \frac{\frac{N}{n} \sum_{i \in s} [M_i \bar{y}_i + d_i (M_i \bar{x}_i - X_i)]}{\frac{N}{n} \alpha \sum_{i \in s} X_i + (1 - \alpha) X}$$

are particular cases of $\hat{Y}_C$.

4. An example

In an effort to make a decisive choice among different estimators, we consider MU284 population available in Sarndal, Swensson and Wretman (1992, p. 660, Appendix C). This population consists of 284 municipalities (ssu) divided into

50 clusters (fsu) with two positively correlated variables 1985 population ($= y$) and 1975 population ($= x$). To estimate Y , let us consider a two-stage sampling with SRSWOR design of $n = 10$ fsu's followed by $m_i = 2$ ssu's from every i in the first-stage sample.

Efficiencies of some estimators discussed above together with the two-stage ratio estimator $\hat{Y}_R = \sum_{i \in s} M_i \bar{y}_i X / \sum_{i \in s} M_i \bar{x}_i$, and two-stage regression estimator $\hat{Y}_{RG} = \frac{N}{n} \sum_{i \in s} M_i \bar{y}_i - \beta_{yx} \left(\frac{N}{n} \sum_{i \in s} M_i \bar{x}_i - X \right)$ are computed in table 4.1.

Table 4.1: Efficiencies of different estimators

Estimator	$\hat{Y}$	$\hat{Y}_R$	$\hat{Y}_{RG}$	$\hat{Y}_{SR}$	$\hat{Y}_{CR}$	t_{RG}
Percentage efficiency	100	348	465	302	416	595

Thus, it can be seen that, as expected, the cluster specific regression-type estimator t_{RG} leads to a substantial increase in precision over others.

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6. References

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